

TALLER MEDIA, VARIACION Y DESVIACION TIPICA**Presentado Por: Anderson Rene Gomez Aza Grupo: S8A Fecha: 04/04/2024****1 PUNTO****Hallar la desviación media, la varianza y la desviación típica de 10, 13, 6, 28, 31** $X_1 = 6, X_2=10, X_3=13, X_4=28, X_5=31$ Y $P=1/5$

Media: $\mu = 6 * 1/5 + 10 * 1/5 + 13 * 1/5 + 28 * 1/5 + 31 * 1/5$	$\mu = 17,6$
Varianza: $\sigma^2 = 6^2 * 1/5 + 10^2 * 1/5 + 13^2 * 1/5 + 28^2 * 1/5 + 31^2 * 1/5 -$	$\sigma^2 = 100,24$
Desviacion Tipica: $\sigma = \sqrt{\text{varianza}}$	$\sigma = 10,0119928086271$

2 PUNTO**Hallar la desviación media, la varianza y la desviación típica de la series de números siguientes:****A) 2, 3, 6, 8, 11** $X_1 = 2, X_2=3, X_3=6, X_4=8, X_5=11$ Y $P=1/5$

Media: $\mu = 2 * 1/5 + 3 * 1/5 + 6 * 1/5 + 8 * 1/5 + 11 * 1/5$	$\mu = 6$
Varianza: $\sigma^2 = 2^2 * 1/5 + 3^2 * 1/5 + 6^2 * 1/5 + 8^2 * 1/5 + 11^2 * 1/5 - \mu^2$	$\sigma^2 = 10,8$
Desviacion Tipica: $\sigma = \sqrt{\text{varianza}}$	$\sigma = 3,286335345031$

B) 12, 6, 7, 3, 15, 10, 18, 5 $X_1 = 3, X_2=5, X_3=6, X_4=7, X_5=10, X_6=12, X_7=15, X_8=18$ Y $P=1/8$

Media: $\mu = (3 + 5 + 6 + 7 + 10 + 12 + 15 + 18) * 1/8$	$\mu = 9,5$
Varianza: $\sigma^2 = ((3^2 + 5^2 + 6^2 + 7^2 + 10^2 + 12^2 + 15^2 + 18^2) * 1/8) - \mu^2$	$\sigma^2 = 23,75$
Desviacion Tipica: $\sigma = \sqrt{\text{varianza}}$	$\sigma = 4,87339717240448$

3 PUNTO**Hallar la varianza y la desviación típica de 5, 7, 6, 5, 10, 13, 6, 8, 7, 9, 9, 10, 5, 8** $X_1=5, X_2=5, X_3=5, X_4=6, X_5=6, X_6=7, X_7=7, X_8=8, X_9=8, X_{10}=9, X_{11}=9,$
 $X_{12}=10, X_{13}=10, X_{14}=13$ Y $P=1/14$

Media: $\mu = (5 + 5 + 5 + 6 + 6 + 7 + 7 + 8 + 8 + 9 + 9 + 10 + 10 + 13) * (1/14)$	$\mu = 7,71428571428571$
Varianza: $\sigma^2 = ((5^2 + 5^2 + 5^2 + 6^2 + 6^2 + 7^2 + 7^2 + 8^2 + 8^2 + 9^2 + 9^2 + 10^2 + 10^2 + 13^2) * (1/14)) - \mu^2$	$\sigma^2 = 4,97302857142857$
Desviacion Tipica: $\sigma = \sqrt{\text{varianza}}$	$\sigma = 2,23159136044214$

4 PUNTO**Hallar la varianza y la desviación media de las notas de taller de matemáticas****3.8, 3.7, 4.5, 3.1, 0.0, 3.1, 2.9, 1.9, 2.6, 3.8, 3.8, 0.0, 4.0, 0.5, 4.6, 3.3, 1.7, 4.5, 3.7, 4.9, 2.0, 3.5, 4.4, 3.5, 4.0, 4.1, 2.7, 3.6, 3.3, 3.0, 3.6, 3.5, 4.4, 3.9, 4.3, 4.2, 4.0, 3.5**

$X_1=0.0, X_2=0.0, X_3=0.5, X_4=1.7, X_5=1.9, X_6=2.0, X_7=2.6, X_8=2.7, X_9=2.9, X_{10}=3.0,$
 $X_{11}=3.1, X_{12}=3.1, X_{13}=3.3, X_{14}=3.3, X_{15}=3.5, X_{16}=3.5, X_{17}=3.5, X_{18}=3.5, X_{19}=3.6,$
 $X_{20}=3.6, X_{21}=3.7, X_{22}=3.7, X_{23}=3.8, X_{24}=3.8, X_{25}=3.8, X_{26}=3.9, X_{27}=4.0, X_{28}=4.0,$
 $X_{29}=4.0, X_{30}=4.1, X_{31}=4.2, X_{32}=4.3, X_{33}=4.4, X_{34}=4.4, X_{35}=4.5, X_{36}=4.5, X_{37}=4.6,$
 $X_{38}=4.9$ y $P = 1/38$

Media: $\mu = (0.0 + 0.0 + 0.5 + 1.7 + 1.9 + 2.0 + 2.6 + 2.7 + 2.9 + 3.0 + 3.1 + 3.1 + 3.3 + 3.3 + 3.5 + 3.5 + 3.5 + 3.5 + 3.6 + 3.6 + 3.7 + 3.7 + 3.8 + 3.8 + 3.8 + 3.9 + 4.0 + 4.0 + 4.0 + 4.1 + 4.2 + 4.3 + 4.4 + 4.4 + 4.5 + 4.5 + 4.6 + 4.9) * (1/38)$	$\mu = 3,31315789473684$
Varianza: $\sigma^2 = (((0,0^2 + 0,0^2 + 0,5^2 + 1,7^2 + 1,9^2 + 2,0^2 + 2,6^2 + 2,7^2 + 2,9^2 + 3,0^2 + 3,1^2 + 3,1^2 + 3,3^2 + 3,3^2 + 3,5^2 + 3,5^2 + 3,5^2 + 3,5^2 + 3,6^2 + 3,6^2 + 3,7^2 + 3,7^2 + 3,8^2 + 3,8^2 + 3,8^2 + 3,9^2 + 4,0^2 + 4,0^2 + 4,0^2 + 4,1^2 + 4,2^2 + 4,3^2 + 4,4^2 + 4,4^2 + 4,5^2 + 4,5^2 + 4,6^2 + 4,9^2) * (1/38)) - (3.32^2)) - \mu^2$	$\sigma^2 = 1,32944210526316$
Desviacion Tipica: $\sigma = \sqrt{\text{varianza}}$	$\sigma = 1,15325625946708$